ARE 106, MON APR 6

SOME NOTES

① SIMPLE REGRESSION — POP / — Sample

vs

② MULTIPLE REGRESSION — POP / — Sample

① Simple - only 1 x variable on RHS

$$y = \beta_1 + \beta_2 x + e \quad - \text{PRF}$$

$$y = b_1 + b_2 x + \hat{e} \quad - \text{SRF}$$

x = INCOME

y = EXPENDITURE or SPENDING

② MULTIPLE REGRESSION

– more than 1 x variable on RHS

$$y = \beta_1 + \beta_2 x_2 + \beta_3 x_3 + \dots + e$$

$$y_i = b_1 + b_2 x_2 + b_3 x_3 + \dots + e$$

LINEARITY – what is it?

① LINEAR in VARIABLES

* LINEAR in PARAMETERS

KEY →

① EXPONENTS MUST BE RAISED to 1

$\beta_2^1 = \beta_2$ (OK)

$\beta_2^2 = \beta_2^2$ (NOT OK)

(2) NO CROSS-PRODUCT TERMS OF PARAMETERS

$\beta_2 \cdot \beta_3$ (NOT OK)

Example 1

(A) $y = \beta_1 + \beta_2 x + e$

LIN - LIN

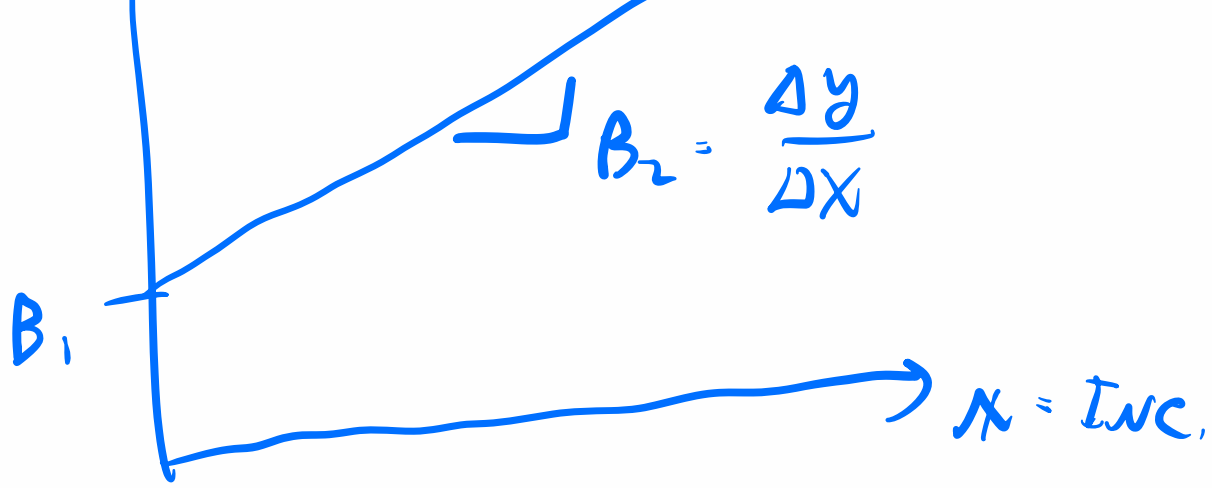
Linear in Variables

Linear in Parameters *

we can use OLS to estimate Parameter values

Since $\boxed{y = \beta_1 + \beta_2 x + e}$ is Linear in Variables

SPENDING = y



$\boxed{\text{NEXT}}$ - calculate ELASTICITY

$$y = B_1 + B_2 x$$

① slope $\rightarrow \frac{dy}{dx} = B_2 = \frac{\Delta y}{\Delta x}$

② ELASTICITY

$$E = \frac{dy}{dx} \cdot \frac{\bar{x}}{\bar{y}}$$

$$E = B_2 \cdot \frac{\bar{x}}{\bar{y}}$$

x^{-1}

(b) $y' = B_1 + B_2 \left(\frac{1}{x}\right) + e$

① NOT LINEAR in VARIABLES

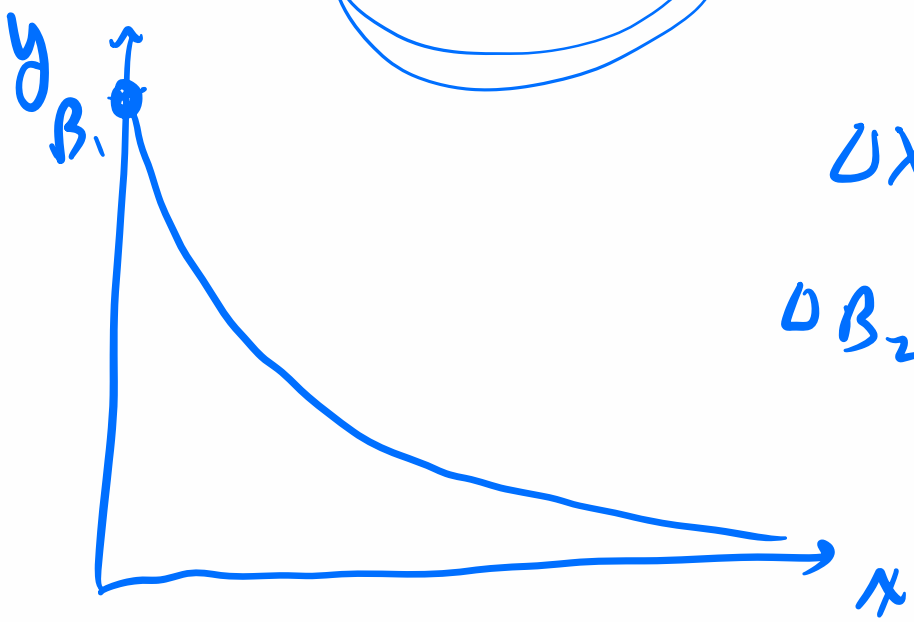
② LINEAR in PARAMETERS

∴ we can use OLS for estimat purposes.

$$y = B_1 + B_2\left(\frac{1}{x}\right)$$

$B_1 > 0$

$B_2 > 0$



$\Delta X \uparrow \rightarrow$

$\Delta B_2\left(\frac{1}{x}\right) \rightarrow \downarrow$

$\uparrow \Delta x$

$$y = B_1 + B_2 \frac{1}{x}$$

$B_2 x^{-1}$

$-1 B_2 x^{-2}$

① $$\frac{dy}{dx} = -B_2 \frac{1}{x^2}$$

④ ELASTICITY

$$\frac{dy}{dx} \cdot \frac{x}{y}$$

$$E = -B_2 \left(\frac{1}{x^2}\right)\left(\frac{x}{y}\right)$$

$$E = -B_2 \left(\frac{1}{xy}\right)$$

(F) $y = B_1 + B_2 B_3 x + e$

(G) $y = B_1 + B_2^2 x + e$

(F), (G)

NOT LINEAR in PARAMETERS.

Thus, we cannot use OLS on These equations

NOTE

① ECONOMIC MODEL

$$y = \beta_1 + \beta_2 x$$

② ECONOMETRIC EQ

$$y = \beta_1 + \beta_2 x + e$$

NOTE

$$E(e) = 0$$

$$\rightarrow Var(e) = \sigma^2$$